[redacted]

ABSTRACT

[redacted] innovation in this area by offering secure document verification methods and serving as a vital tool against document fraud and abuse. The documentation and verification process for academic certificates may be made safe, [redacted]

RESEARCH OBJECTIVES

- TO1: [redacted]
- TO2:
- TO3:
- TO4:
- TO5:
- TO6:

[redacted]

The figure above depicts The peer-to-peer relationship enabled by the decentralized identity model returning people to direct, private connections secured by public/private key cryptography. [3]

Threat Modeling

We have opted for a model [redacted] which is a well established threat model. Now, we will present a variety of the possible security risks which we might face.

Requirement Analysis

- Functional requirements
- Security Requirement
- Privacy Requirement

The SSI Model

[redacted]

To fully imple[redacted]
following fou[redacted]

- **Establishin**[redacted]
  an issuer, v[redacted]
  requests a [redacted]
- **Providing** [redacted]
  give a VC t[redacted]
- **Storing VC**[redacted]
  verifiable c[redacted]
  public/priv[redacted]
- **Requesting**[redacted]
  connection[redacted]

System Architecture

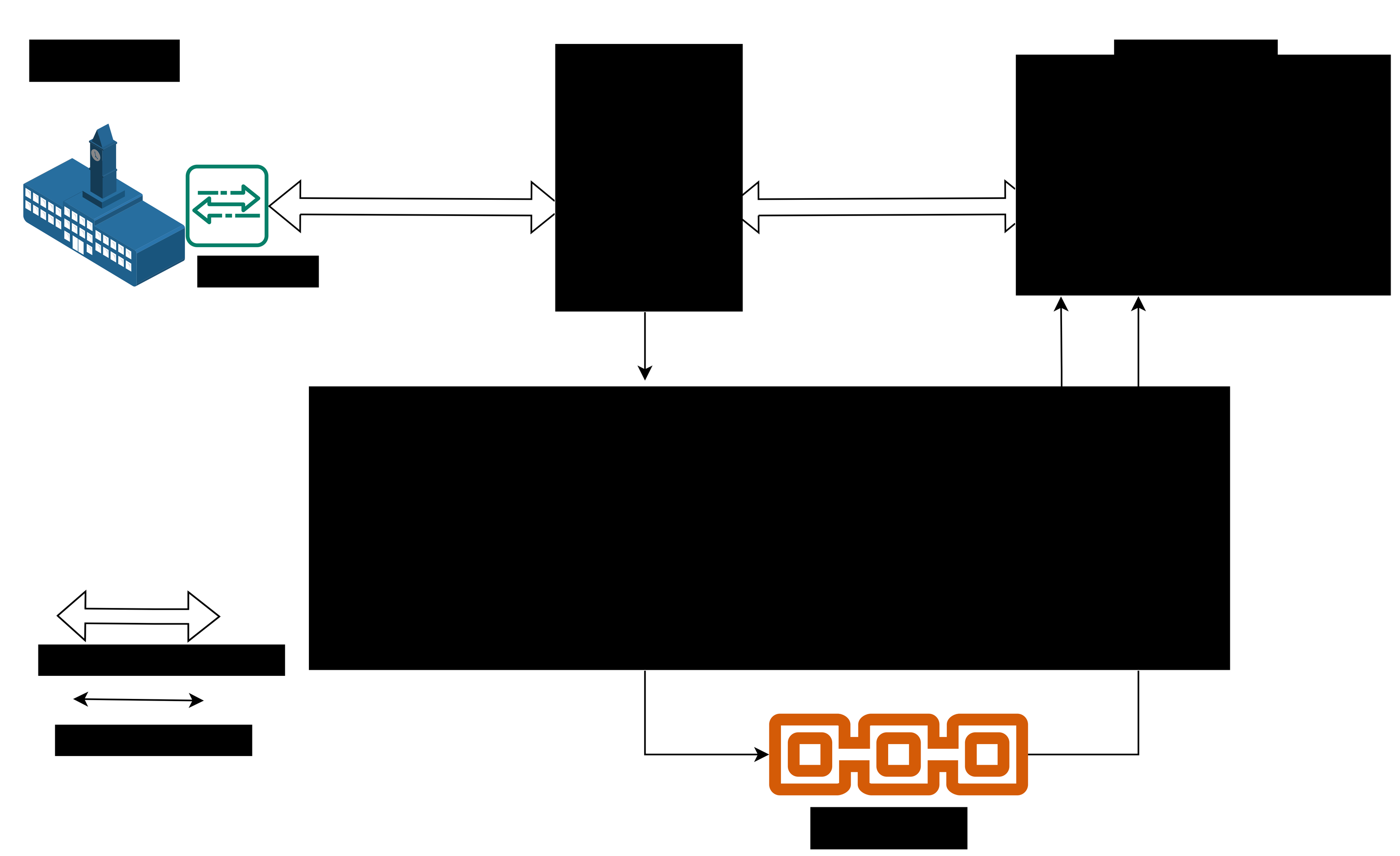

Figure 1. Architecture of Academic Document Verification Using SSI and Blockchain

- The issuer issues SSI-va[redacted] interactions. The issuer [redacted] and private key [1].
- The distributed ledger w[redacted] issuer's agent. Blockcha[redacted]
- A firm or organization m[redacted] construct the proof of V[redacted] verifier's agent. The veri[redacted] blockchain [3].
- The intermediary agent [redacted] the issuer, or both. In a [redacted] messages intended for t[redacted] network [2].

Preliminary Analysis

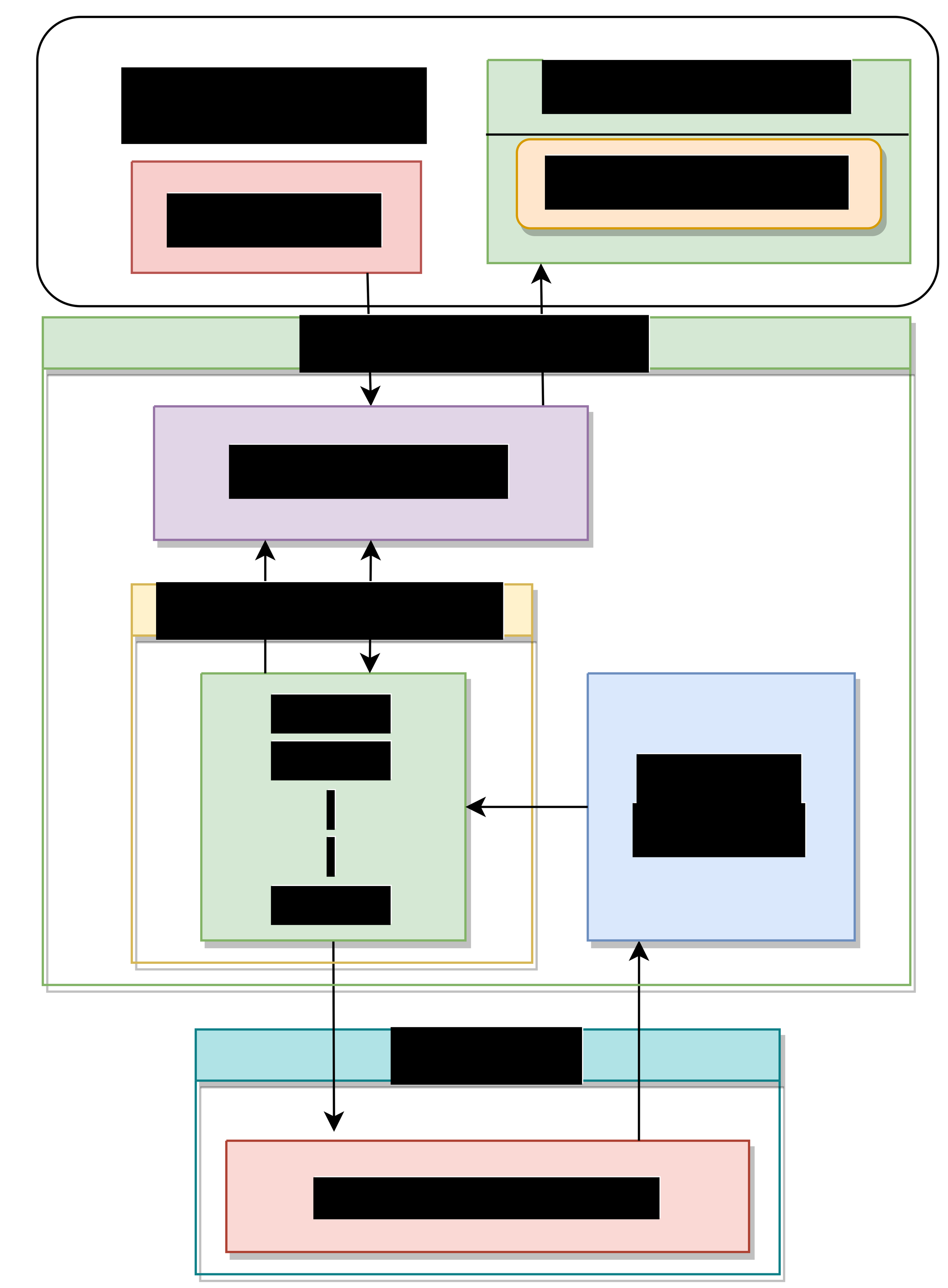

Figure 2. The Architecture of Hyperledger Aries Cloud Agent

References

[1] Verifiabl[redacted]

[2] Md Sad[redacted] Ssi4web[redacted] *Lecture*[redacted]

[3] Alex Pre[redacted] *Self-sove*[redacted] Manning[redacted]